NeuroText

EEG representation learning for sentence retrieval

Can EEG retrieve sentence meaning beyond matched negative controls?

ZuCo 1.0 + 2.0 | Statistical ridge · CNN · SSL · frozen LaBraM

Ziad Elsayed | University project

Waveform and transfer findings with validation-only TRT follow-ups.

Define the prediction task

Fixed-pool retrieval with unseen readers

1. Encode a sentence-length EEG recording.
2. Predict its location in a frozen sentence-embedding space.
3. Rank unique candidate sentences by cosine similarity.
4. Evaluate Recall@1/5/10 and mean reciprocal rank (MRR).

Above random ranking alone is insufficient: real EEG must beat matched controls.

Data preparation and quality audit

Normal reading · 105 ordered channels · resampled to 200 Hz

ZuCo 1.0

12 readers

3,356 prepared observations

3,349 after duration QC

ZuCo 2.0

18 readers

5,911 prepared observations

5,884 after duration QC

19-channel LaBraM adaptation is a separate cohort: 3,345 / 5,838 observations.

One shared, memory-conscious implementation

Windows + Colab clients use the same Python package

LOAD

Lazy signal arrays

Audited manifests

REPRESENT

Statistics / CNN

SSL / frozen LaBraM

ALIGN

Train-only scaling

Validation selection

RANK

MiniLM targets

Cosine retrieval

Local hardware: RTX 3050 with 4 GB VRAM; lazy loading limits memory use.

Shared code and documented inputs make weak results inspectable and reproducible.

Controls make the central comparison

Hold out subjects; fit preprocessing on training observations only

Real EEG: original signals and true sentence pairing.

Gaussian: matched per-channel scale, independently generated noise.

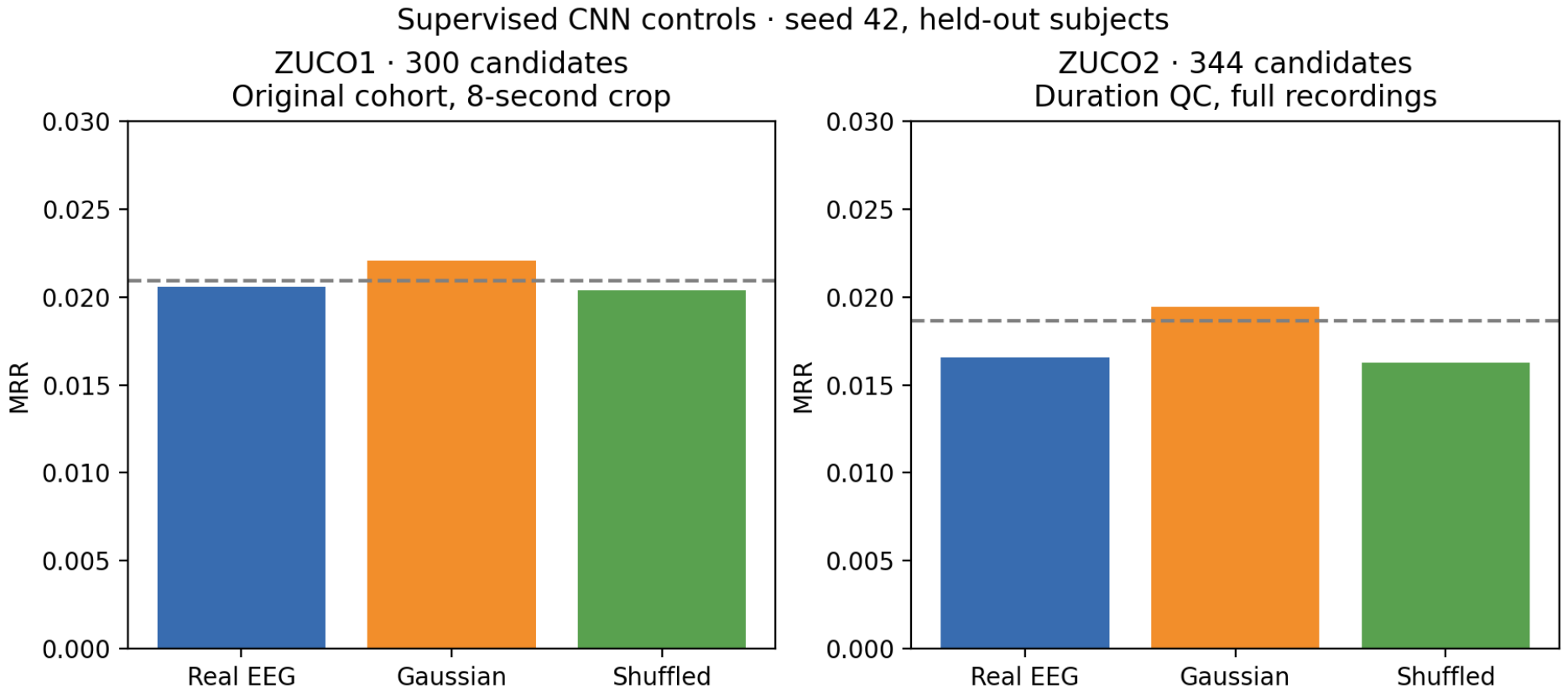
Shuffled: separately trained model with incorrect sentence pairing.

Compare within the same cohort, split, seed and candidate pool.

Single-split differences are descriptive; no statistical significance is claimed.

Supervised CNN: Gaussian matches top-1 hits

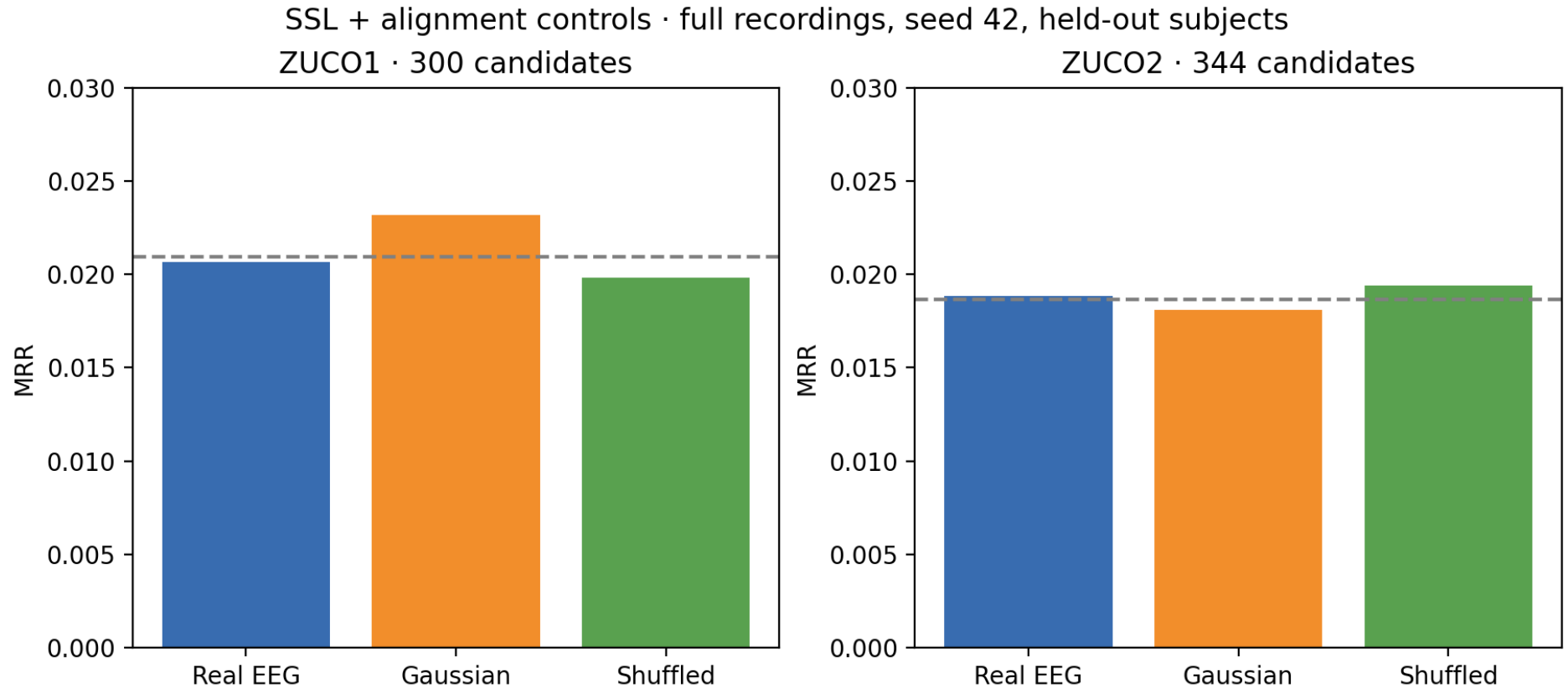
Seed 42 · ZuCo1: 592 queries / 300 candidates · ZuCo2: 984 / 344



Gaussian MRR exceeds real EEG in both datasets.

SSL: reconstruction does not establish decoding

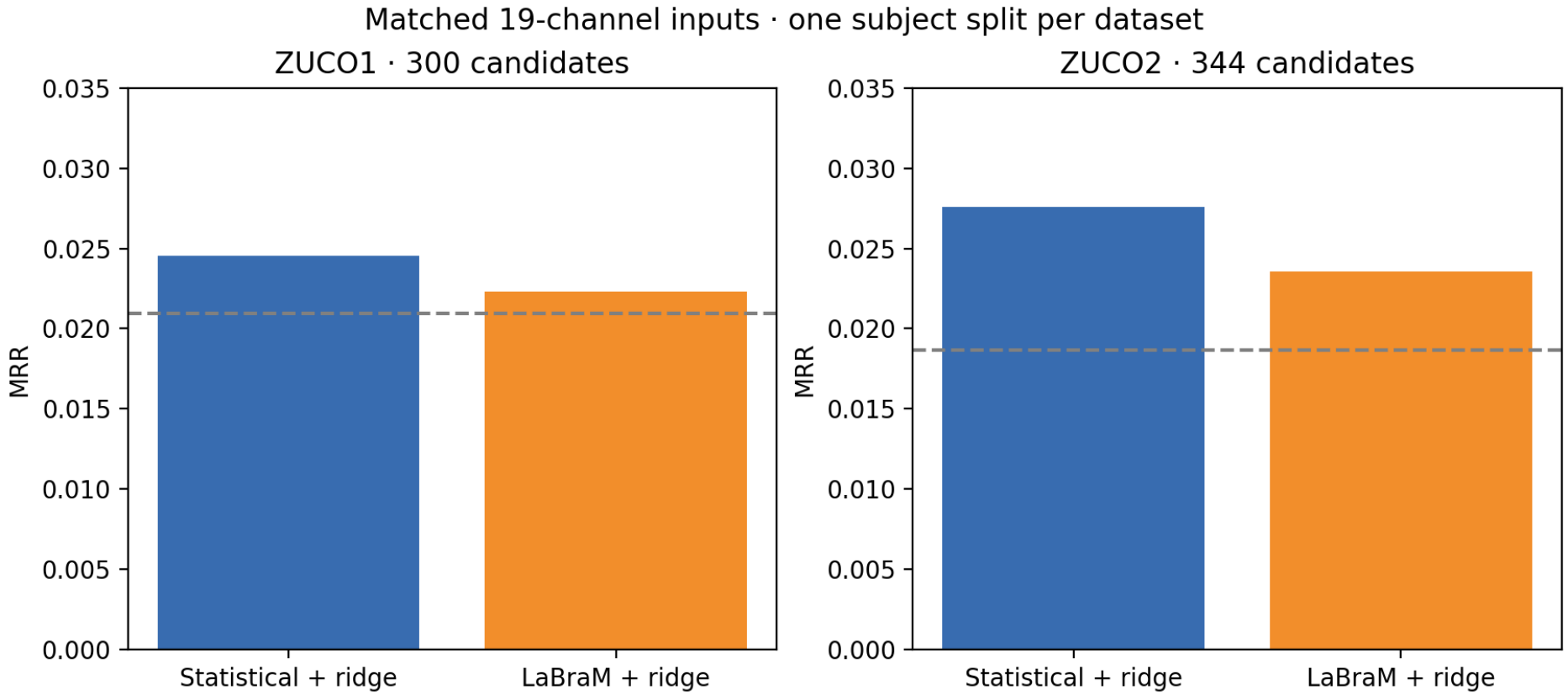
Full recordings · duration QC · Gaussian control covers both learned stages



No consistent retrieval advantage for real EEG over the controls.

Frozen LaBraM against matched statistical features

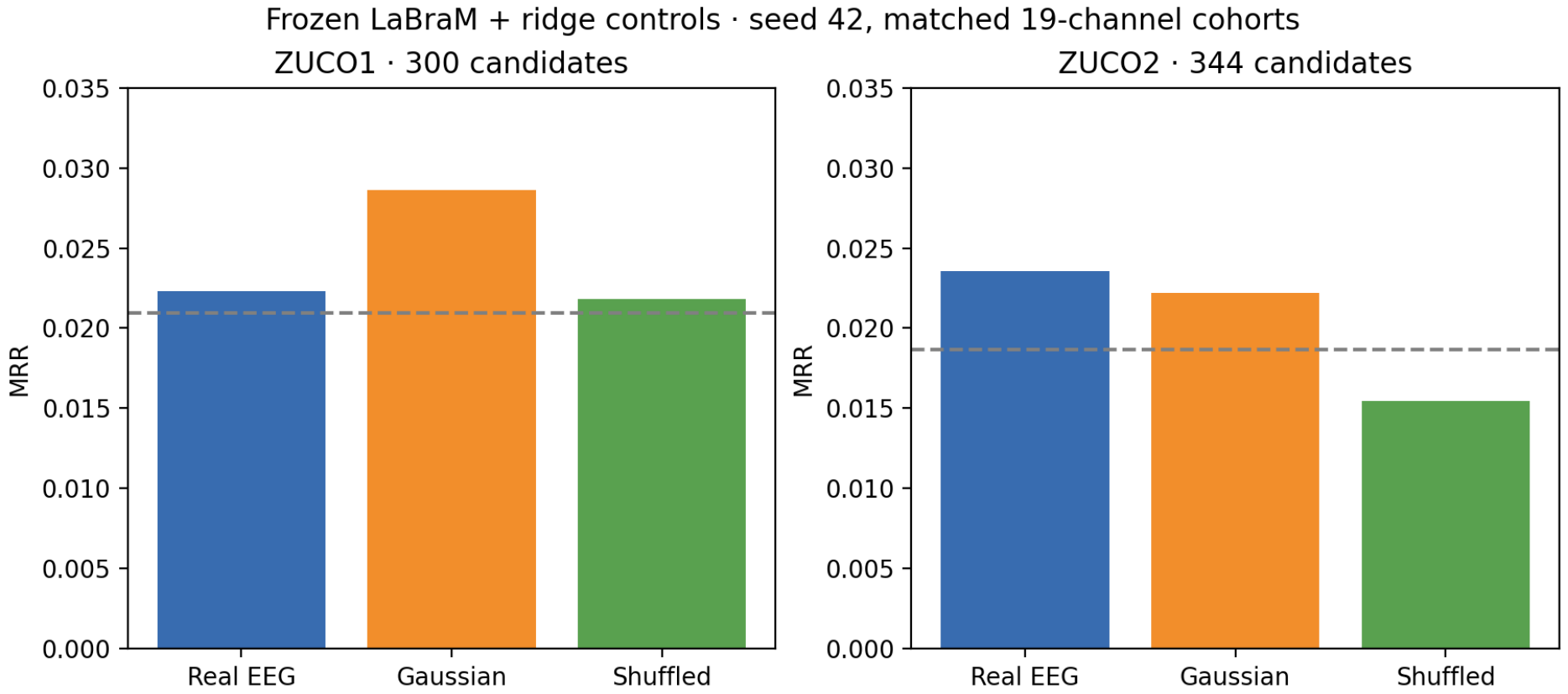
Identical adapted signals · 19 mapped electrodes · whole-second support



Statistical features have higher top-1 and MRR on both reported splits.

LaBraM controls limit the interpretation

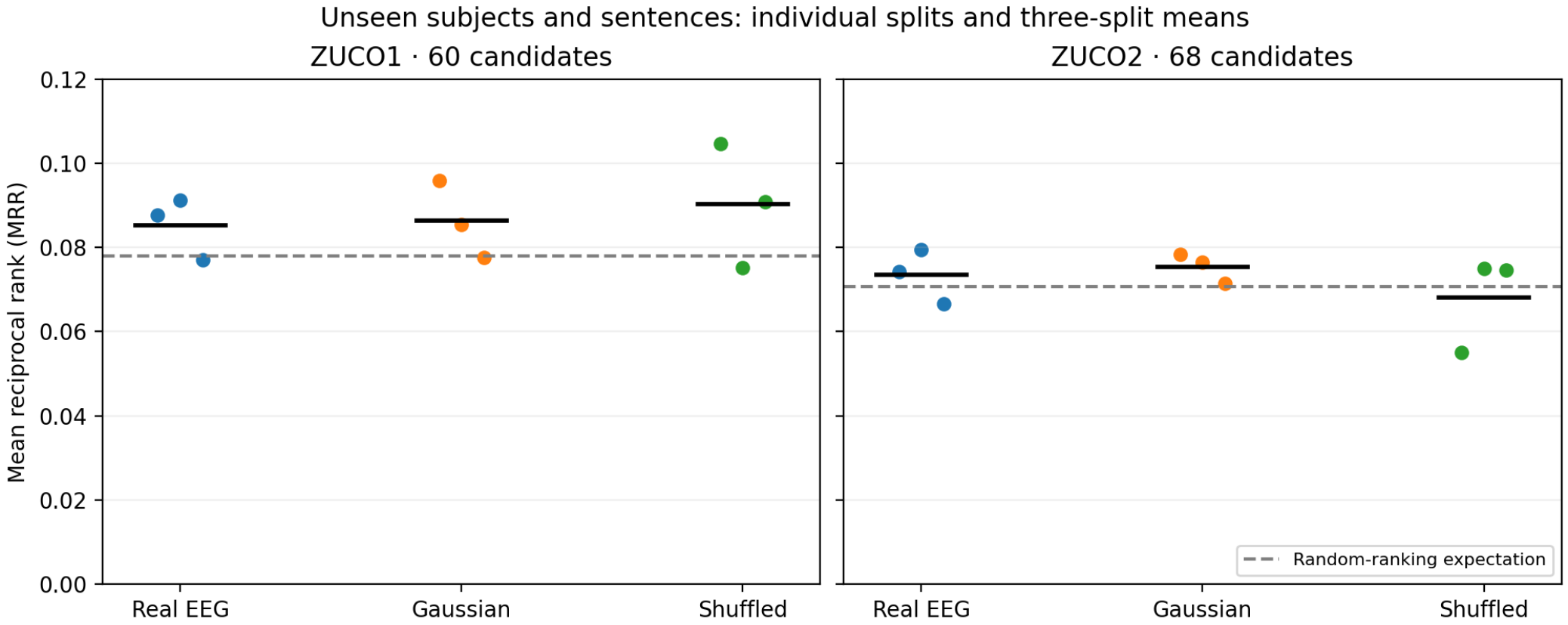
Frozen backbone · separately fit ridge models · same adapted cohort and split



No consistent real-EEG superiority under this exploratory adaptation.

Unseen subjects and unseen sentence content

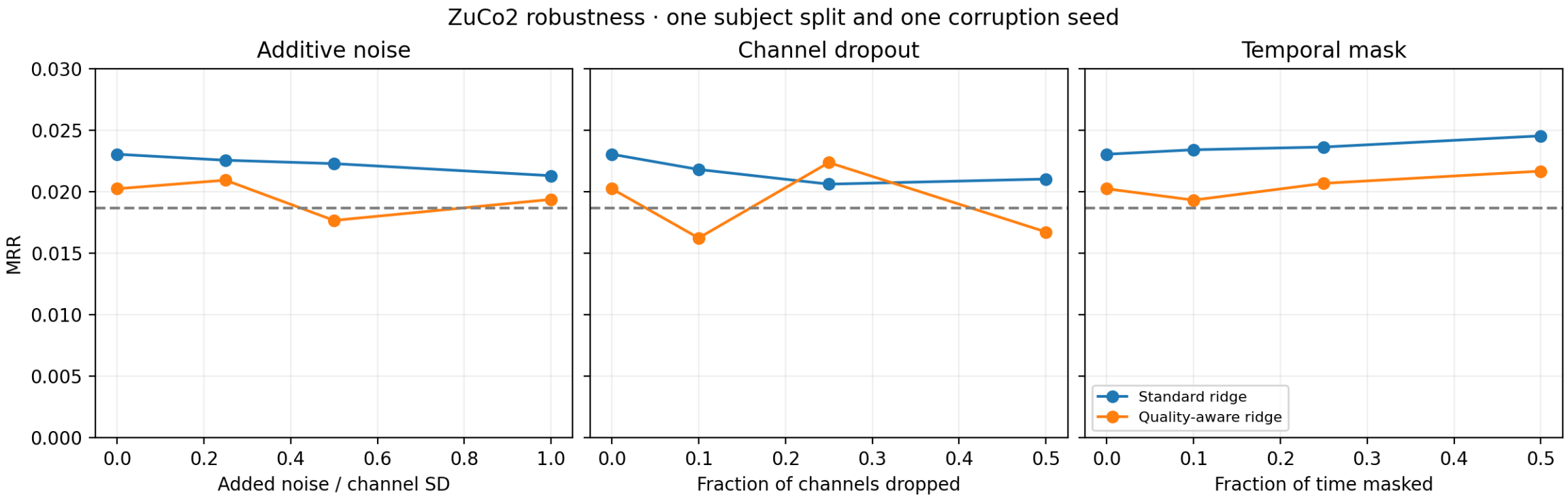
Ridge · seeds 42, 43, 44 · 60 candidates in ZuCo1 / 68 in ZuCo2



Real EEG exceeds Gaussian MRR on only one of three seeds in each dataset.

Quality weighting shows no consistent benefit

ZuCo2 ridge · 984 queries / 344 candidates · one split and corruption seed



Quality-aware MRR is lower in 8 of 9 nonzero corruption conditions.

Published work: compare protocols before scores

Generation, classification and retrieval measure different tasks

EEG-to-text papers often report BLEU / ROUGE for language generation.

LaBraM originally reports classification tasks.

Our task reports Recall@K / MRR for a fixed sentence pool.

The original AAAI 2022 repository documents an evaluation correction.

Primary-source protocol review is complete; competitive numerical ranking is unsupported.

Exploratory transfer uses source-only selection

18 wider-grid evaluations · both directions · seeds 42, 43, 44

Train signal normalization and ridge on source training observations only.

Select alpha by source-validation MRR across seven fixed values.

Compare real EEG with separately trained Gaussian and shuffled controls.

Participant independence across releases remains unresolved.

50 shared texts remain in the target pools; strict transfer is still unexecuted.

Wider-grid results: ZuCo1 to ZuCo2

5,884 queries · 344 candidates · the same target cohort at every seed

Seed	Condition	Alpha	Source val MRR	Recall@1	Target MRR
42	Real	100	0.028249	0.340%	0.018732
42	Gaussian	100	0.025700	0.306%	0.019507
42	Shuffled	100000	0.020376	0.272%	0.018177
43	Real	1000	0.024743	0.255%	0.019005
43	Gaussian	0.1	0.027825	0.493%	0.020673
43	Shuffled	0.1	0.027011	0.238%	0.017791
44	Real	0.1	0.022323	0.238%	0.017851
44	Gaussian	10	0.020144	0.238%	0.017859
44	Shuffled	100	0.022480	0.238%	0.018772

Real EEG MRR is below Gaussian MRR on all three seeds.

Wider-grid results: ZuCo2 to ZuCo1

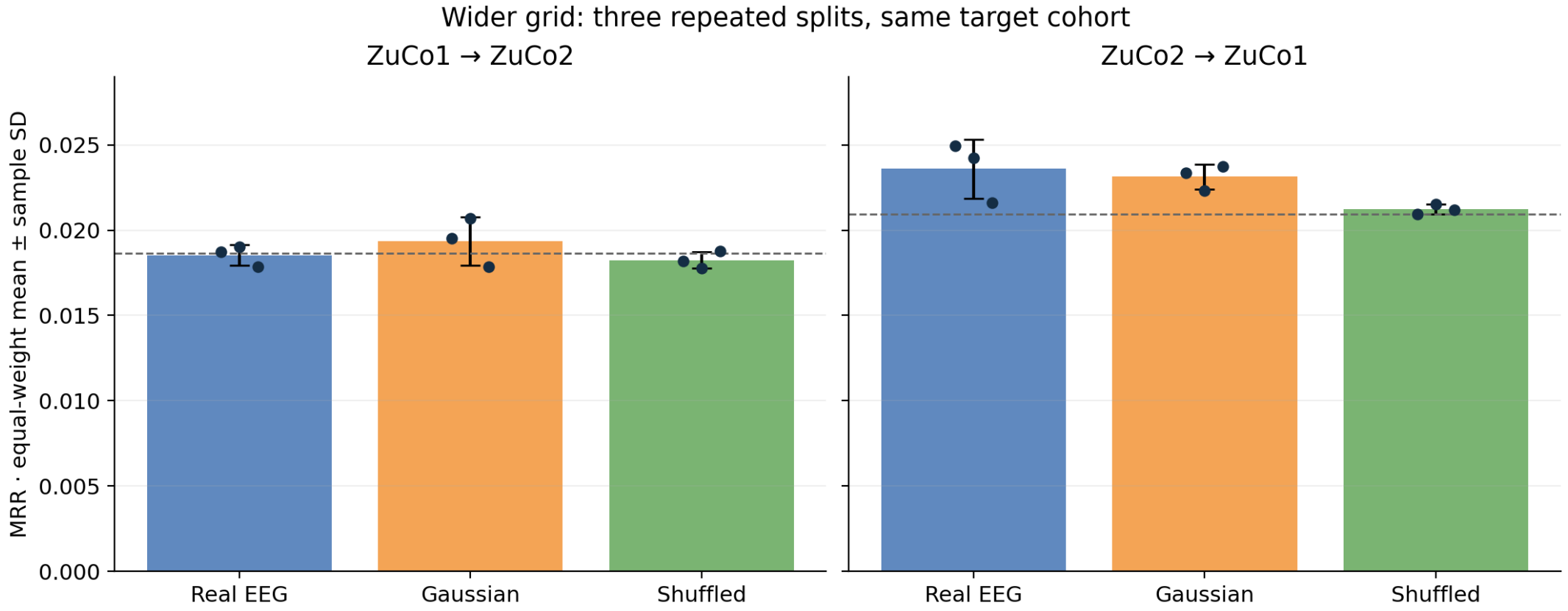
3,349 queries · 300 candidates · the same target cohort at every seed

Seed	Condition	Alpha	Source val MRR	Recall@1	Target MRR
42	Real	1000	0.026690	0.627%	0.024933
42	Gaussian	1000	0.024125	0.448%	0.023354
42	Shuffled	100000	0.019260	0.299%	0.020941
43	Real	1000	0.024824	0.508%	0.024221
43	Gaussian	1	0.025861	0.358%	0.022323
43	Shuffled	1000	0.020172	0.388%	0.021521
44	Real	100000	0.021306	0.388%	0.021621
44	Gaussian	10	0.025757	0.567%	0.023748
44	Shuffled	1	0.016905	0.418%	0.021185

Real EEG MRR exceeds Gaussian on seeds 42 and 43; the sign reverses on seed 44.

Wider-grid means remain close to controls

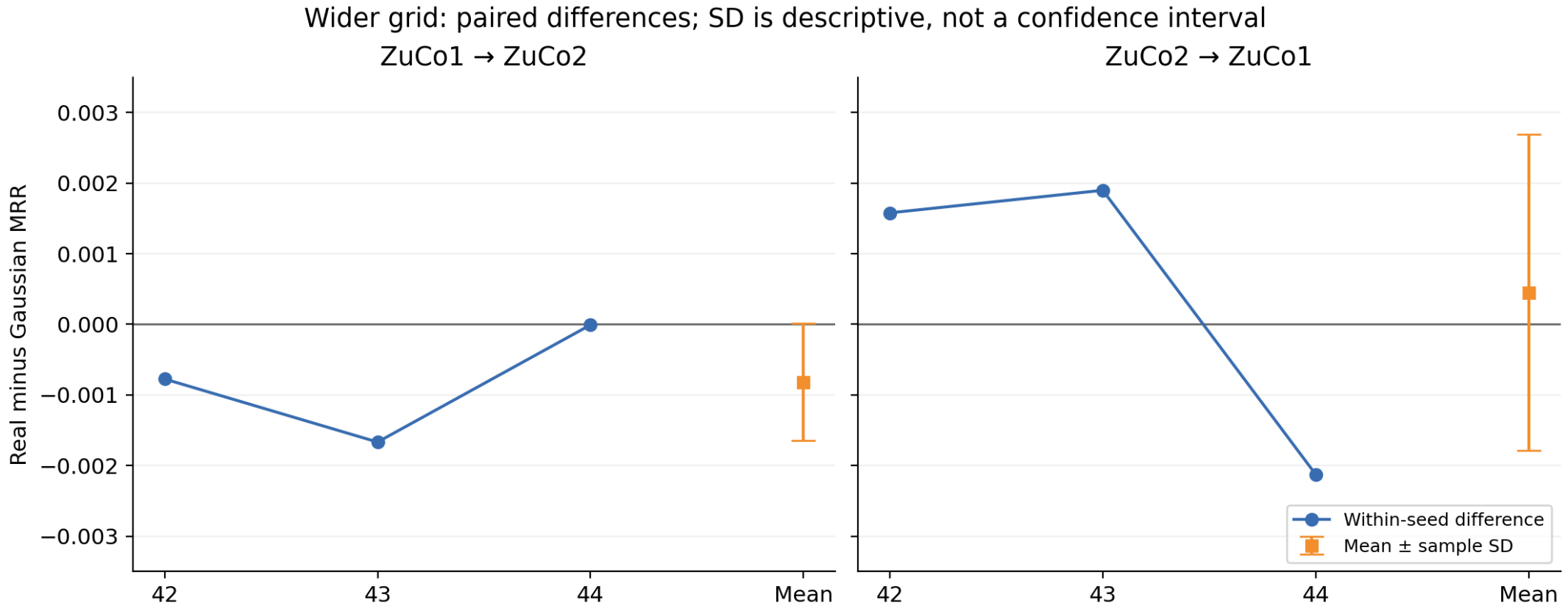
Equal-weight seed means · sample SD · individual seed points



Repeated splits reuse target participants and queries; SD is descriptive.

Real-minus-Gaussian differences change by seed

Pair within direction and seed before calculating mean and sample SD



Forward ΔMRR −0.000817 ± 0.000831; reverse +0.000450 ± 0.002237.

Original and wider grids are separate experiments

Seed 42 only · original grid ends at 100 · wider grid ends at 100,000

Direction	Condition	Original α / MRR	Wider α / MRR
1 to 2	Real	100 / 0.018732	100 / 0.018732
1 to 2	Gaussian	100 / 0.019507	100 / 0.019507
1 to 2	Shuffled	1 / 0.019001	100000 / 0.018177
2 to 1	Real	10 / 0.022941	1000 / 0.024933
2 to 1	Gaussian	100 / 0.024007	1000 / 0.023354
2 to 1	Shuffled	10 / 0.021036	100000 / 0.020941

Only the wider grid has seeds 42–44; do not pool the grids as extra repeats.

Archive and recorded software validation

139 passed · 2 optional skips · 0 deselected

Full configs, candidate keys, split digests and evaluation JSON are archived.

Two official-weight tests skip because EEG_LABRAM_CACHE is unset.

Integration skip: no network download during unit tests.

Extraction skip: optional official weights on synthetic fixtures.

The previously excluded 513 MiB MATLAB test was included and passed.

Recorded validation is preserved; consolidation launches no experiments.

These configurations do not show reliable retrieval

Matched controls and repeated splits limit the interpretation

These configurations do not demonstrate reliable retrieval beyond controls.

Forward real MRR trails Gaussian on all three seeds.

Reverse differences change sign, with only a small positive mean.

Participant independence remains unresolved; strict transfer is unexecuted.

The findings do not establish that EEG contains no semantic information.

Validation-only TRT spectral pilot

TRT pools eight 105-channel word vectors into sentence features.

The cohort has 2,245 training observations and 512 validation queries.

All validation scores rank the same 298 sentence candidates.

Counts and missingness require explicit structural controls.

Gaze-assisted source features retain timing and preprocessing limitations.

Validation-only pilot favors structural controls

Condition	Val R@1	Val R@5	Val R@10	Val MRR
Real TRT	12.305%	32.227%	41.406%	0.226986
Matched Gaussian	10.742%	23.633%	28.906%	0.177208
Class shuffled	0.391%	2.148%	3.711%	0.022676
Count / duration	17.773%	40.625%	51.367%	0.292012
Zero / missingness	16.406%	40.234%	52.344%	0.281028
Sentence ridge	0.586%	2.734%	4.102%	0.026320

Count / duration validation MRR 0.292012 uses no measured TRT amplitudes.

Validation-only incremental comparison

All three models share 62 structural features and one frozen transform.

Combined models append 840 real or matched Gaussian TRT values.

Both combined models search 196 settings, including zero EEG weight.

Post-scaling weights preserve an exact structural-only candidate.

196 settings = 2 EEG imputers x 2 EEG scalers x 7 alphas x 7 weights

Validation-only incremental results

Model	Val R@1	Val R@5	Val R@10	Val MRR	EEG weight
Structure	17.773%	40.625%	51.367%	0.292012	0
Structure + real	17.383%	41.016%	51.562%	0.292310	0.01
Structure + Gaussian	17.773%	40.820%	51.172%	0.292156	0.01

Val delta MRR: real - structure +0.000299, real - Gaussian +0.000154

Validation-only reader effects change sign

Reader	Queries	Val real - structure MRR	Val real - Gaussian MRR
ZJS	216	-0.003530	-0.004365
ZKW	296	+0.003092	+0.003452

Real TRT loses two Recall@1 hits overall despite the small pooled MRR increase.

Validation-only results show no consistent EEG gain

The structural baseline retrieves sentence classes from reading structure.

Real TRT adds no consistent benefit across both validation readers.

The 196-setting search and repeated validation use limit inference.

Further tuning and test evaluation are stopped.

Every new score is validation-only. No test features were extracted.

Later pilots: what changed after 30 October

SAVED RESULTS ONLY / No training or evaluation ran for this update / No test-set use

Study	Change tested	Scope	Supported finding
Ordered-word ridge	Four word-position bins vs. mean	CV, then reused validation	One extra hit; controls stronger
Relative amplitude	Channel-relative features, two heads	Training CV	Zero EEG weight selected
Full-sequence GRU	All ordered word steps	Training CV	Gate failed; far below structure
GRU retrieval loss	Full-candidate cross entropy	Training CV	Predictions spread; no better retrieval
Word pretraining	Exported-word targets first	Training CV	No gain over fresh training
N-way analysis	Smaller random candidate pools	Replayed training CV	Real EEG near chance, below missingness

Training CV and reused validation are development scores, not untouched confirmation.

Ordered-word ridge does not beat its controls

REUSED OUTER VALIDATION / ZuCo1 seed 42 / 512 queries / 298 candidates / alpha chosen on training readers

Condition	R@1	R@5	R@10	MRR
Mean real EEG	0.195%	1.758%	3.320%	0.019783
Ordered real EEG	0.391%	1.367%	4.102%	0.022452
Matched Gaussian	0.781%	1.953%	4.102%	0.026921
Word-order shuffled	0.586%	1.172%	4.688%	0.024464
Class shuffled	0.000%	2.148%	3.516%	0.019378
Missingness only	17.578%	40.625%	51.758%	0.291229
Structure only	17.578%	40.430%	51.367%	0.291369

Order adds one top-1 hit over the mean; Gaussian and order-shuffled score higher on R@1 and MRR.

Relative amplitudes: both heads select zero EEG weight

TRAINING-READER CV ONLY / 8 readers / 2,245 observations / 300 candidates

Condition	Embedding R@1	Embedding MRR	Known-class R@1	Known-class MRR
EEG only	0.355%	0.020519	0.357%	0.019292
Gaussian only	0.696%	0.025876	0.395%	0.022132
Structure only	16.113%	0.280271	20.909%	0.400506
Structure + real EEG	16.113%	0.280271	20.909%	0.400506
Structure + Gaussian	16.113%	0.280271	20.909%	0.400506

Combined models equal structure alone. The 20.909% known-class score is structure, not EEG.

GRU: the loss change spreads predictions, not retrieval

TRAINING-READER CV ONLY / 8 readers / 2,245 observations / 300 candidates / epoch 8

Condition	Cosine R@1	Cosine MRR	Cross-entropy R@1	Cross-entropy MRR
Real EEG	0.542%	0.023048	0.529%	0.020781
Mean EEG	0.401%	0.022454	0.363%	0.019303
Matched Gaussian	0.404%	0.021327	0.261%	0.018486
Class shuffled	0.358%	0.021432	0.174%	0.020120
Word-order shuffled	0.441%	0.022068	0.265%	0.018809
Missingness only	0.358%	0.021191	0.449%	0.023184
Structure only	16.113%	0.280271	16.113%	0.280271

Distinct top-1 choices rise from 6.375 to 53 per reader; real beats Gaussian in only 3/8 readers.

Word pretraining adds no sentence-retrieval gain

TRAINING-READER CV ONLY / 8 readers / 2,245 observations / 300 candidates / sentence scores

Condition	R@1	R@5	R@10	MRR
Pretrained real EEG	0.395%	1.382%	2.986%	0.020149
Fresh, same sentence updates	0.529%	1.553%	2.903%	0.021318
Fresh, equal total updates	0.348%	1.579%	2.890%	0.019855
Pretrained Gaussian	0.216%	1.023%	2.295%	0.017447
Pretrained class shuffled	0.578%	1.824%	3.720%	0.023627
Pretrained missingness	0.898%	2.846%	5.086%	0.030513
Structure only	16.113%	40.287%	52.053%	0.280271

Real EEG beats fresh same-sentence training in 0/8 readers and missingness in 0/8 readers.

Word-task scores are a separate quantity

WORD CLASSES, NOT SENTENCES / Training-reader CV / Macro across 8 readers / Text is a target only

Condition	R@1	R@5	R@10	MRR
Pretrained real EEG	3.008%	10.046%	15.943%	0.068325
Pretrained Gaussian	3.002%	9.727%	15.503%	0.067224
Pretrained missingness	3.333%	14.504%	21.836%	0.088084
Pretrained class shuffled	0.067%	0.199%	0.324%	0.003046

Missingness alone scores higher overall; only 1/8 readers favor real EEG on word Recall@1.

Smaller candidate pools change difficulty, not the model

EXPECTED ACCURACY / Replayed training-reader CV / Uniform random distractors / Not a new cohort

N	Real EEG	Gaussian	Class shuffled	Missingness	Structure	Chance
2	49.21%	49.20%	50.23%	53.83%	90.44%	50.00%
5	19.53%	18.81%	20.26%	23.77%	75.06%	20.00%
10	9.71%	8.80%	10.27%	12.83%	62.81%	10.00%
20	4.74%	4.05%	5.28%	7.03%	51.36%	5.00%
50	1.81%	1.42%	2.29%	3.24%	37.88%	2.00%
100	0.89%	0.65%	1.27%	1.87%	28.53%	1.00%
300	0.39%	0.22%	0.58%	0.90%	16.11%	0.33%

Two-way real EEG is 49.21%, below 50% chance. Structure's 90.44% is annotation-derived, not EEG.

Project brief: what the evidence covers

Coverage of the seven brief steps / Done means executed and reported, not that decoding succeeded

Brief step	Evidence	Status
1. Data and splits	ZuCo 1.0 and 2.0 prepared; QC cohorts 3,349 and 5,884; reader-disjoint train, validation and test.	**Done**
2. Preprocessing	105 ordered channels, 200 Hz, train-fit normalizers; upstream session processing is inherited.	**Done**
3. Representations	Frozen adapted LaBraM, ZuCo reconstruction SSL, supervised CNN, statistical baselines; no useful gain.	**Done**
4. Decoding and noise control	Fixed-pool sentence retrieval; Gaussian and shuffled controls; controlled negative result.	**Done**
5. Quality and robustness	ZuCo2 ridge only, one split and corruption seed; quality weighting lower in 8 of 9 conditions.	**Partial**
6. New subjects and datasets	Unseen readers tested; cross-release transfer exploratory; strict transfer not run.	**Partial**
7. Evaluation and analysis	Methods compared on clean data; degraded conditions cover ridge only; limitations documented.	**Partial**

Open items: strict transfer, and degraded-condition coverage beyond ZuCo2 ridge.

Current conclusion: a controlled negative result

All scopes considered / Scores from different scopes are not ranked against each other

Supported

Reproducible pipeline with matched controls and verified source pairing.

Strong retrieval from annotation-derived counts and durations, separate from EEG.

Not supported

A useful, consistent EEG gain beyond the strongest controls.

Verified participant-independent transfer between ZuCo releases.

One validated final accuracy.

Needed

A revised, falsifiable hypothesis with a frozen plan.

Independent readers and stimuli for generalization claims.

Untouched evaluation, not more tuning on this development history.

This does not show that EEG lacks language information; these configurations did not establish it.